## Chapter 5

# Switch Mode DC-AC Inverters

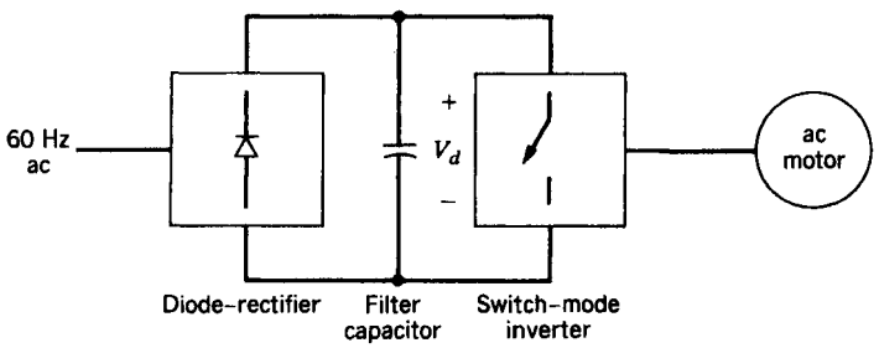
**Instructor: Le Hoa Nguyen, Ph.D.**

# Introduction

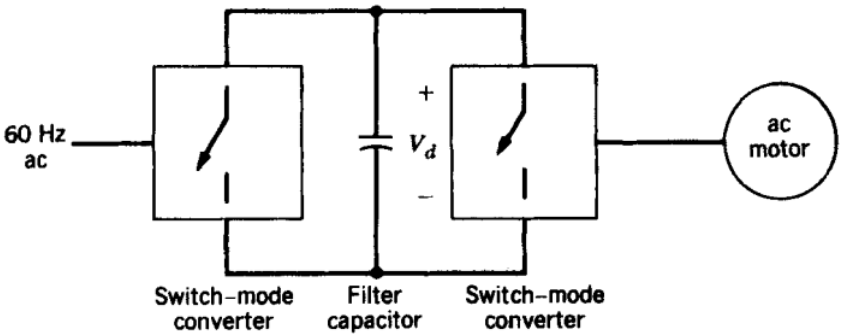
Objective: To convert a DC source (voltage/current) to a sinusoidal AC output whose magnitude and frequency can both be controlled

Applications:

- AC motor drives
- Uninterruptible Power Supplies (UPS)




One-directional converter (inverter)



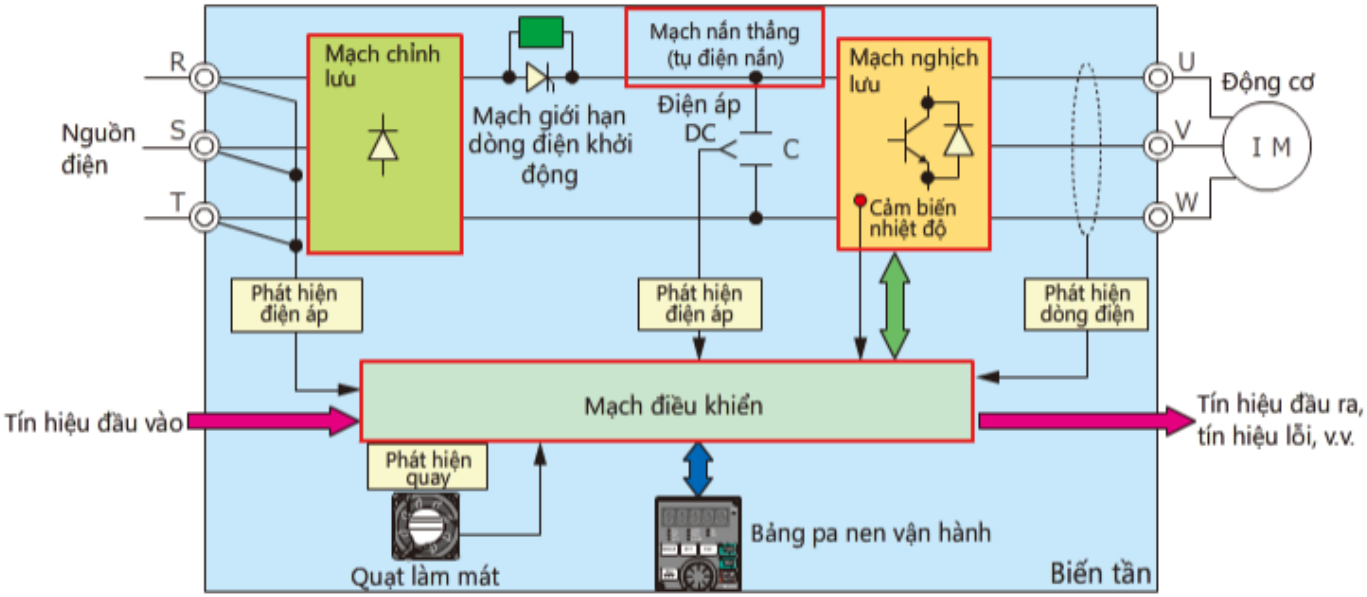

Bi-directional converter

Categories of voltage source inverters (VSI):

- Pulse-width modulated inverter
- Square-wave inverter
- Single-phase inverter with voltage cancellation

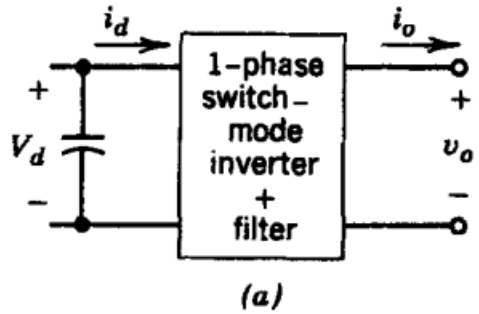
# Introduction

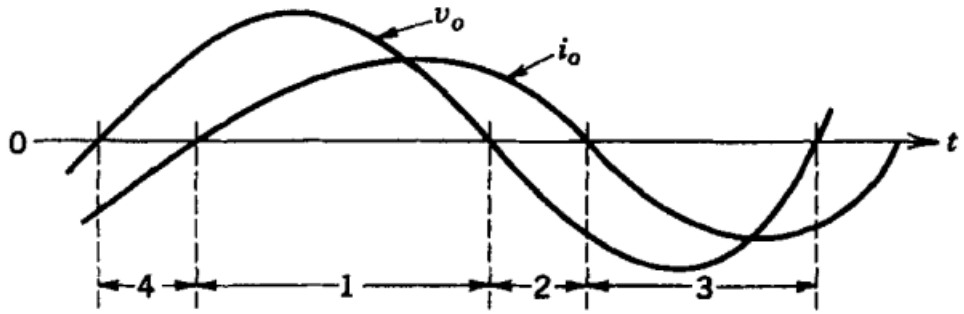


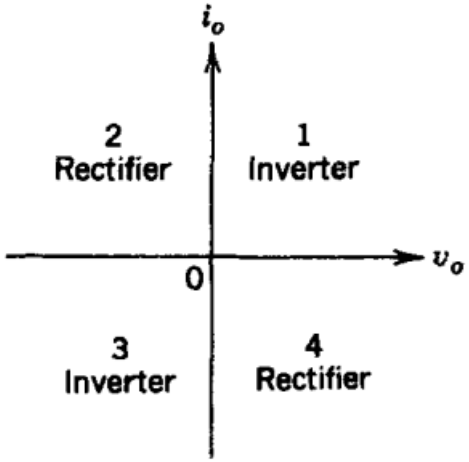




| Tên mạch | Vai trò |
|---|---|
| Mạch chỉnh lưu | Chuyển AC sang DC. |
| Tụ điện nắn | Nắn điện áp DC được chuyển đổi trong Mạch chỉnh lưu |
| Mạch nghịch lưu | Đảo dòng DC thành AC tại tần số xác định bằng mạch điều khiển. |
| Mạch điều khiển | Nhận lệnh từ tín hiệu đầu vào và gửi lệnh đến Mạch nghịch lưu.<br>Cung cấp trạng thái Mạch nghịch lưu. |

# Basic concepts of switch-mode inverter



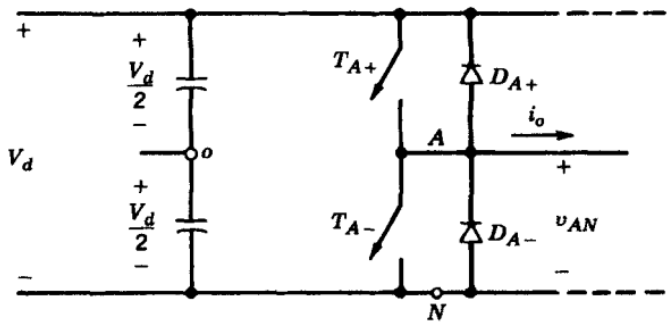



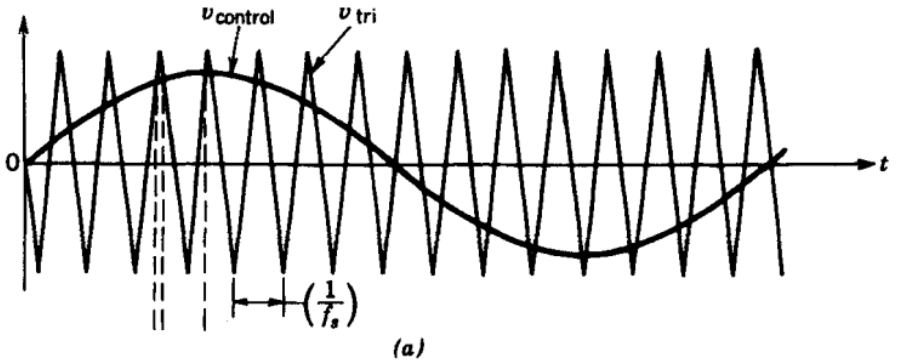



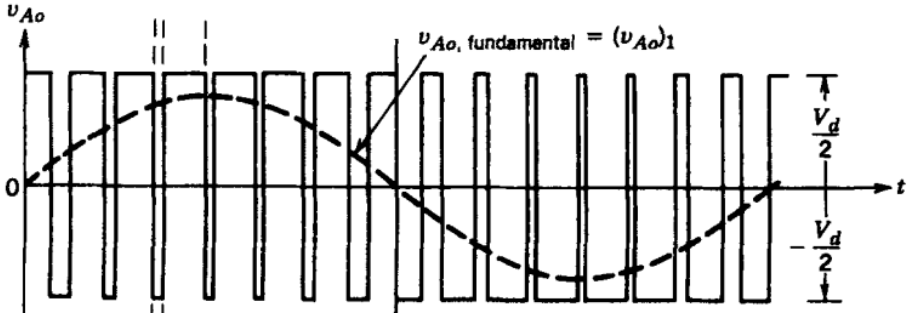

- During intervals 1, 3: $i_0$, $v_0$ are the same sign, hence, $p_0 = i_0 v_0 > 0$, power flows from the dc side to the ac side => inverter mode of operation
- During invertals 2, 4: $i_0$, $v_0$ are the opposite sign, $p_0 = i_0 v_0 < 0$, power flows from the ac side to the dc side => rectifier mode of operation

# Basic concepts of switch-mode inverter

## Pulse-width-modulated (PWM) switching scheme



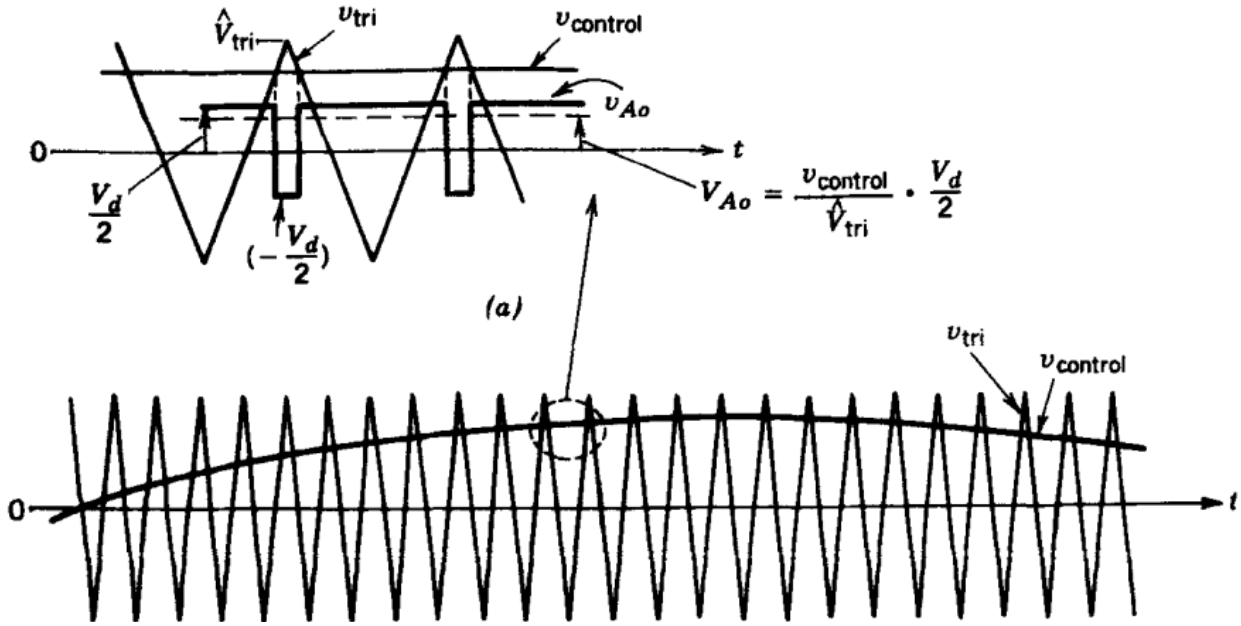




- The freq. of $v_{control}$ ($f_1$) is the desired freq. of the output voltage
- The freq. of trianglar waveform ($v_{tri}$) determines the switching freq. (i.e., $f_s$)
- If $v_{control} > v_{tri}$: $T_{A+}$ is on, $T_{A-}$ is off; $v_{Ao} = V_d/2$
- If $v_{control} < v_{tri}$: $T_{A+}$ is off, $T_{A-}$ is on; $v_{Ao} = -V_d/2$
- Amplitude modulation ratio: $m_a = \dfrac{\hat{V}_{control}}{\hat{V}_{tri}}$
- Frequency modulation ratio: $m_f = \dfrac{f_s}{f_1}$

# Basic concepts of switch-mode inverter

## Pulse-width-modulated (PWM) switching scheme




Control voltage can be assumed constant during a switching time-period

'Moving average' of $v_{A0}$ :

$$V_{A0} = \frac{v_{control}}{\hat{V}_{tri}} \frac{V_d}{2} \quad with \quad v_{control} < \hat{V}_{tri} \qquad (1)$$

where:

$$v_{control} = \hat{V}_{control} \sin \omega_1 t \qquad (2)$$

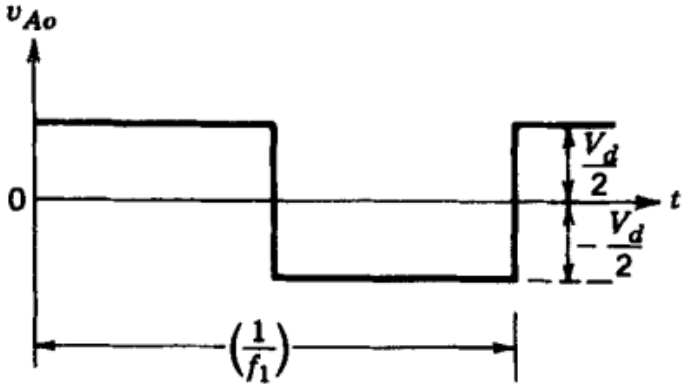
From (1) and (2):

$$V_{A0,1} = \frac{\hat{V}_{control}}{\hat{V}_{tri}} \frac{V_d}{2} \sin \omega_1 t = m_a \frac{V_d}{2} \sin \omega_1 t$$

# Basic concepts of switch-mode inverter

## Square wave switching scheme

- Each switch of the inverter leg is on for one half-cycle ($180^0$) of the desired output frequency




- The peak value of the fundamental frequency component of the output waveform

$$\left(\hat{V}_{A0}\right)_{1,\max} = \frac{4}{\pi}\frac{V_d}{2} = 1.273\frac{V_d}{2}$$

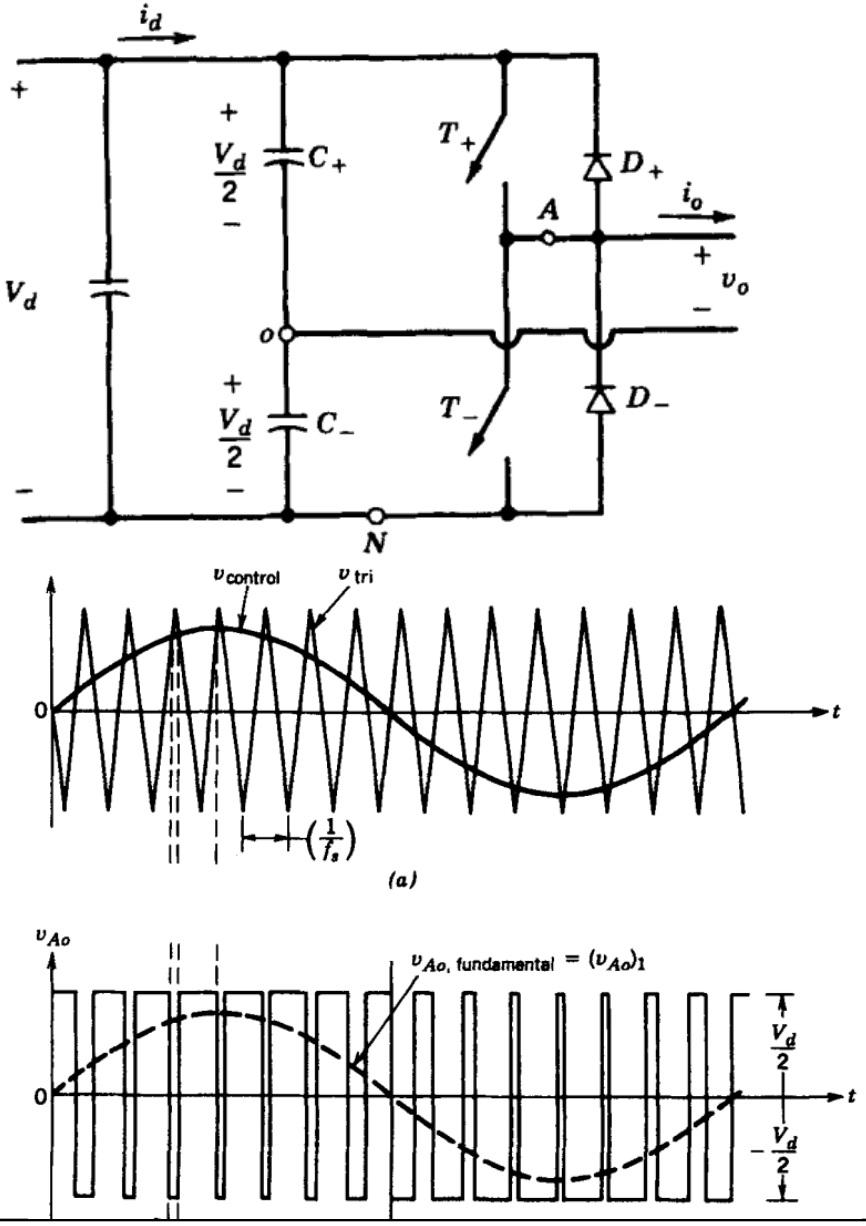
# Half-bridge inverter (single phase)

## PWM with bipolar voltage switching

- Two capactiors are equal and sufficiently large capacitances
- The current between two capacitors divided equally
- When $T_+$ is on, either $T_+$ or $D_+$ conducts depending on the direction of the output current $i_o$ and $i_o$ splits equally between two capacitors
- Similarly, when $T_-$ is on, either $T_-$ or $D_-$ conducts depending on the direction of $i_o$, and $i_o$ splits equally between two capacitors.
- Peak voltage and current rating of the switches

$$V_T = V_d$$

$$I_T = i_{o,peak}$$



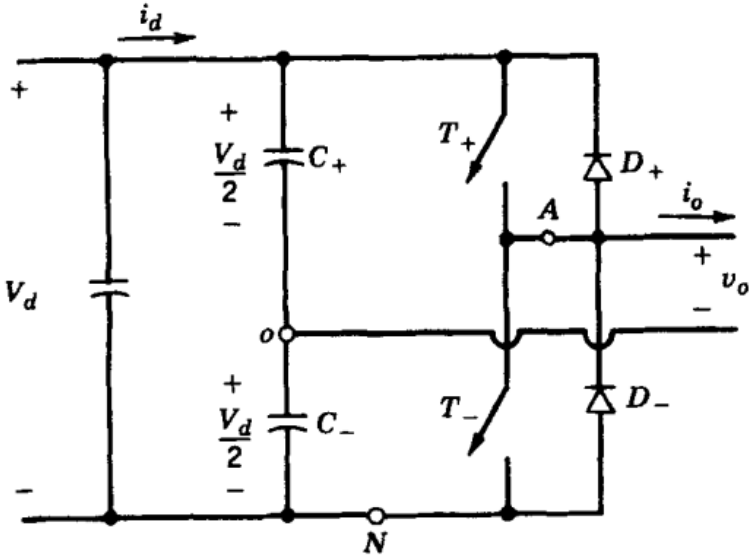
# Half-bridge inverter (single phase)

## PWM with bipolar voltage switching

**Ex**: Consider a half-bridge inverter as shown. The freq. of triangle signal is 1450Hz, the input dc voltage is 600V. Output voltage is sinusoidal voltage with freq. of 50 Hz.

a) Find the freq. modulation ratio, mf

b) Calculate the rms value of the first harmonic output voltage when the amplitude modulation ratio is 0.8



# Full-bridge inverter (single phase)

## PWM with bipolar voltage switching

- Switch pairs ($T_{A+}$, $T_{B-}$), ($T_{A-}$, $T_{B+}$)
- When $T_{A+}$, $T_{B-}$ are on

$$V_{Ao} = +\frac{1}{2}V_d$$

$$V_{Bo} = -\frac{1}{2}V_d$$

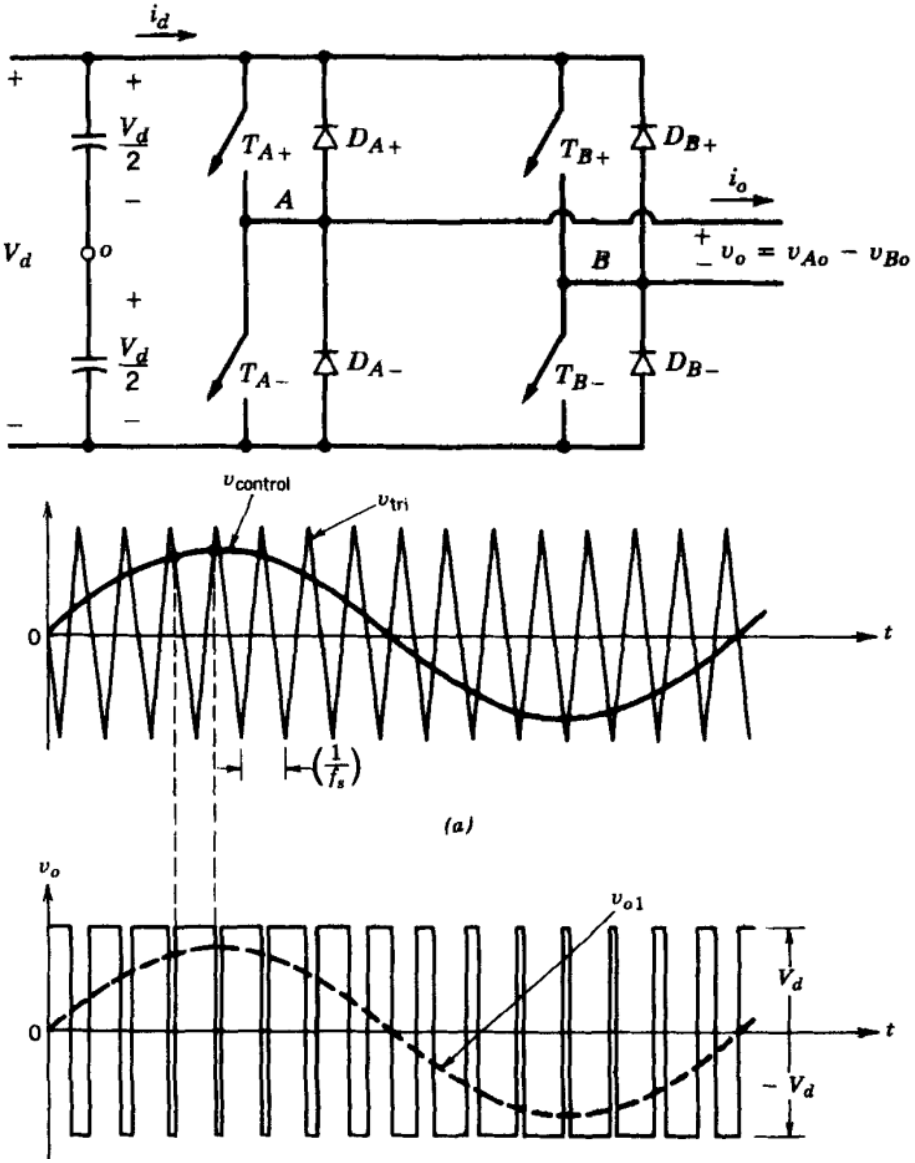
$$v_{Bo}(t) = -v_{Ao}(t)$$

$$\Rightarrow v_o(t) = v_{Ao}(t) - v_{Bo}(t) = 2v_{Ao}(t)$$

The peak of fundamental frequency component of output voltage

$$\hat{V}_{o1} = m_a V_d \qquad m_a \le 1$$

$$V_d < \hat{V}_{o1} < \frac{4}{\pi}V_d \qquad m_a > 1$$



# Full-bridge inverter (single phase)

## PWM with unipolar voltage switching

- Switches in the two legs are controlled separately

Leg A:

$v_{control} > v_{tri}$ $T_{A+}$ is on; $v_{AN} = V_d$

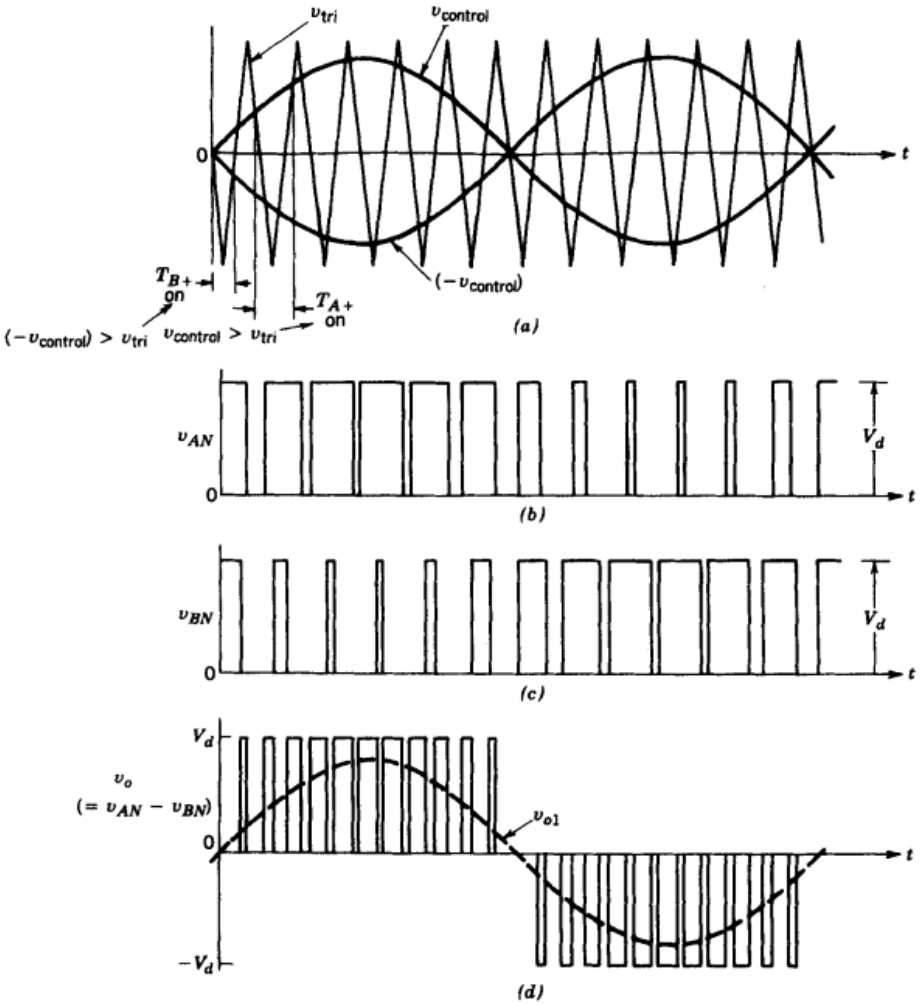
$v_{control} < v_{tri}$ $T_{A-}$ is on; $v_{AN} = 0$

Leg B:

$(-v_{control}) > v_{tri}$ $T_{B+}$ is on; $v_{BN} = V_d$

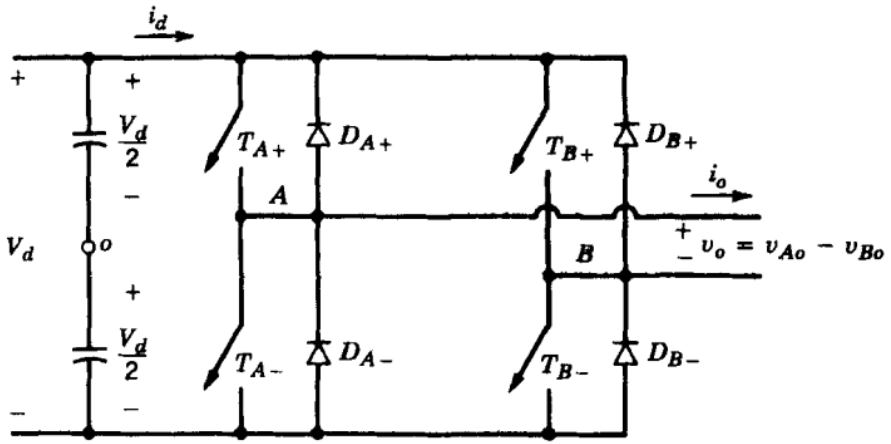
$(-v_{control}) < v_{tri}$ $T_{B-}$ is on; $v_{BN} = 0$

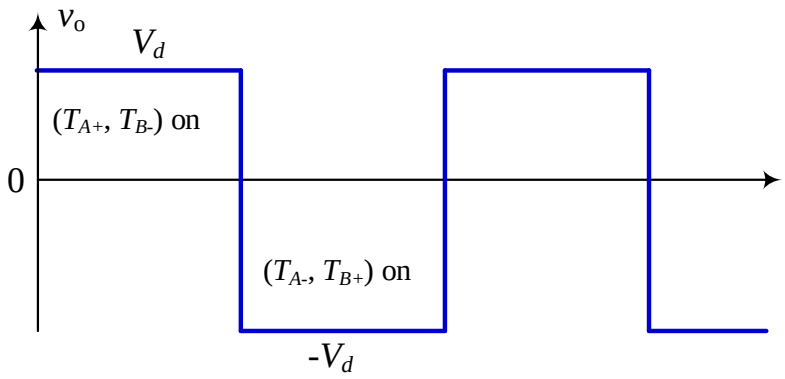
- Same output voltage and magnitude as bipolar switching
- Lower ripple current (about 50%)
- Less harmonic



# Full-bridge inverter (single phase)

## Square-wave operation



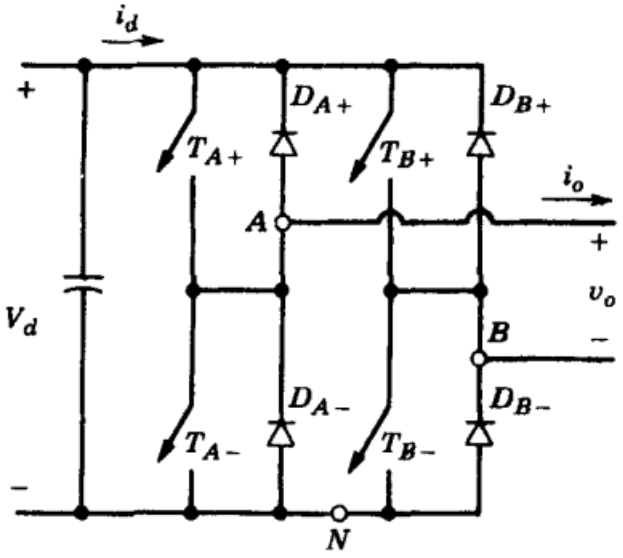

- The switches ($T_{A+}$, $T_{B-}$) and ($T_{A-}$, $T_{B+}$) are operated as two pairs with duty cycle of 0.5.
- The peak of the fundamental frequency component of the output voltage

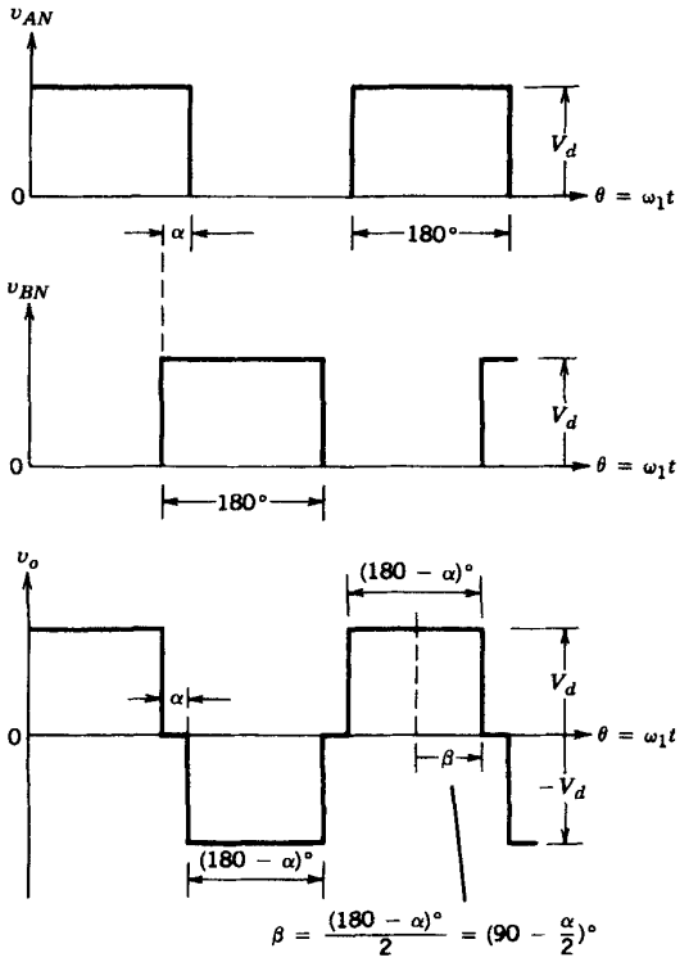
$$\hat{V}_{o1} = \frac{4}{\pi} V_d$$ (output voltage is regulated by controlling the input voltage)

# Full-bridge inverter (single phase)

## Output control by voltage cancellation

- Combination of square-wave switching and PWM with a unipolar voltage switching
- The switches in the two legs are controlled separately (similar to PWM with a unipolar voltage switching); All switches have duty cycle of 0.5 (similar to square-wave switching)




- The overlap angle α can be controlled